Robot Control I

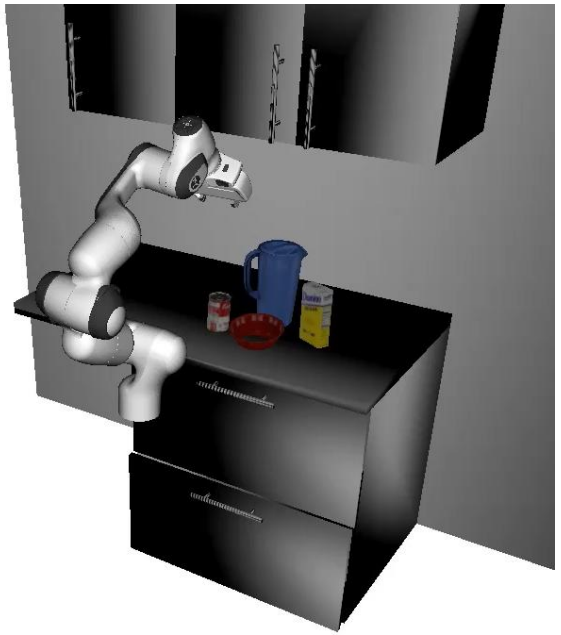
CS 6341 Robotics

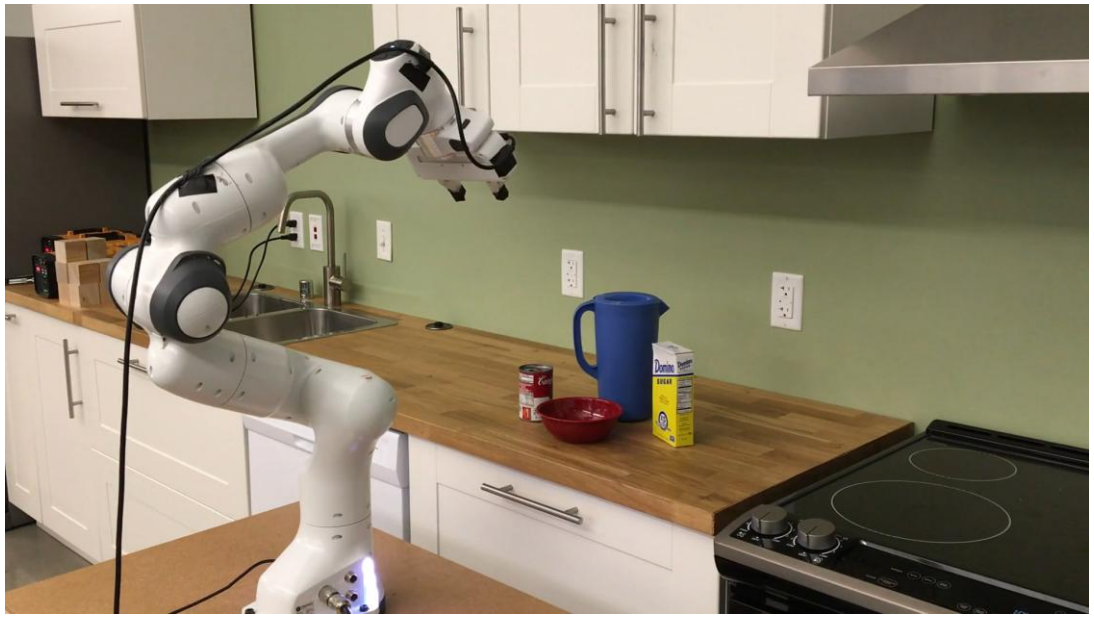
Instructor Yu Xiang

The University of Texas at Dallas

Motion Control

- Goal: follow a given robot trajectory

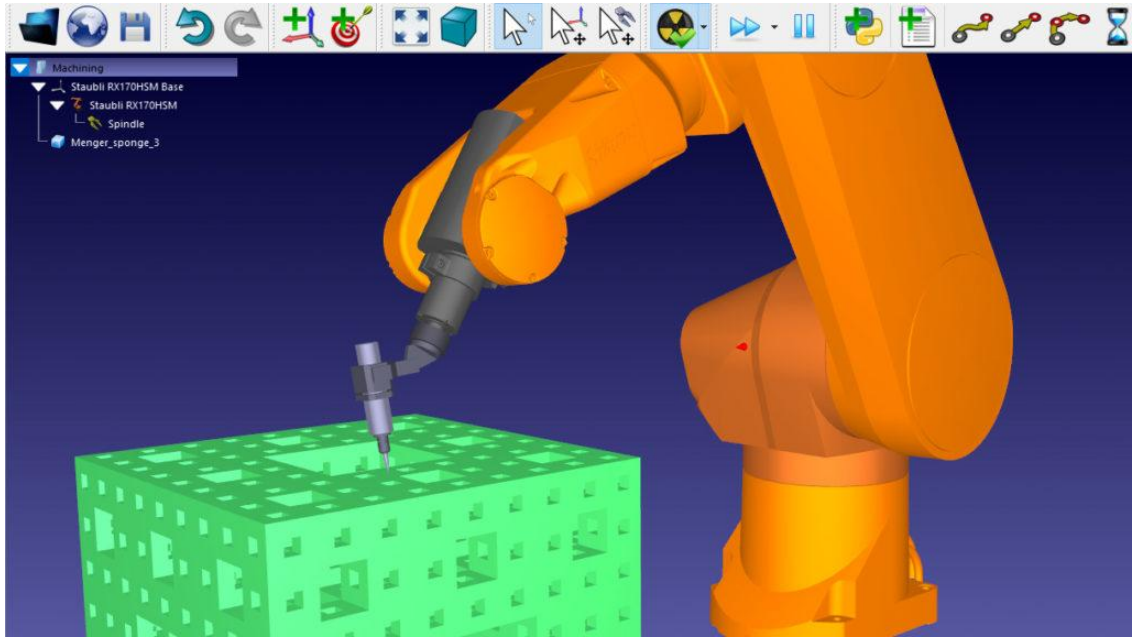




Force Control

- Goal: apply forces and torques to the environments





https://robodk.com/blog/force-control-robot-machining/

Hybrid Motion-Force Control

- Goal: generate both force and motion



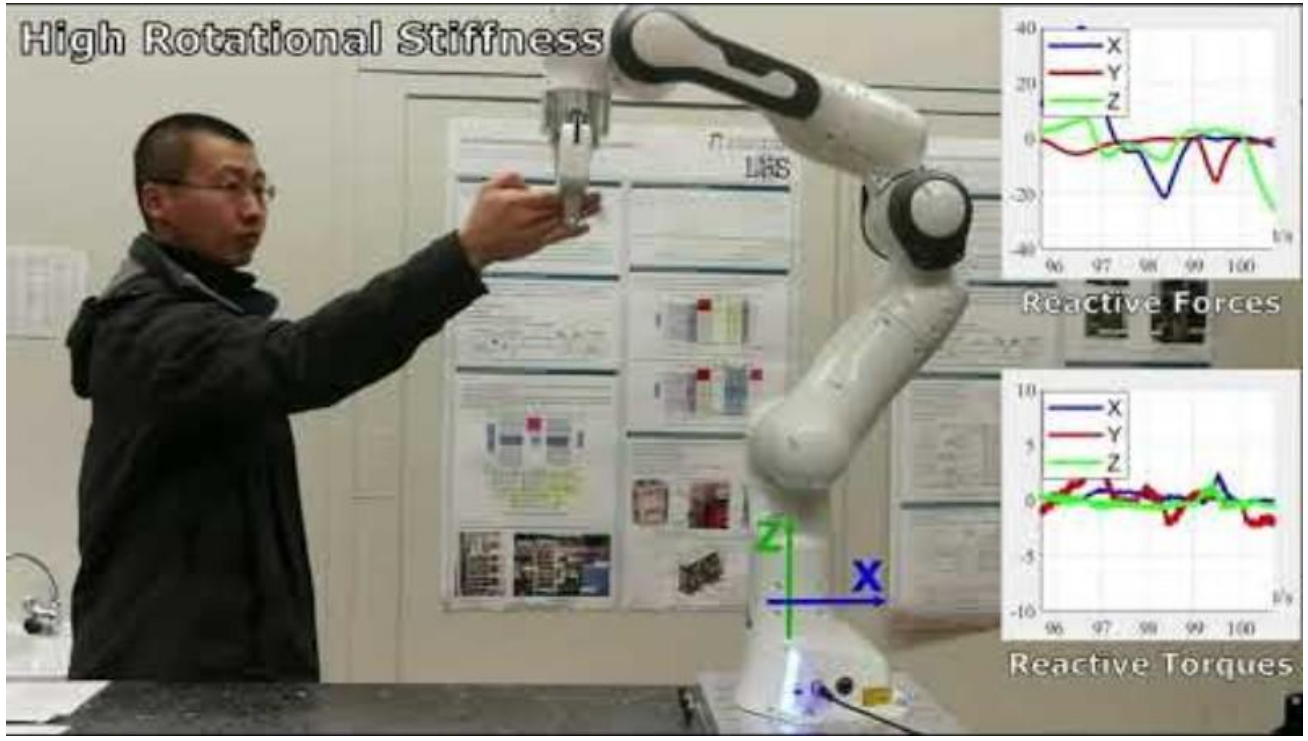
https://youtu.be/9NbwE4PMeyQ

Impedance Control

- Goal: robot end-effector is asked to render particular mass, spring, and damper properties.



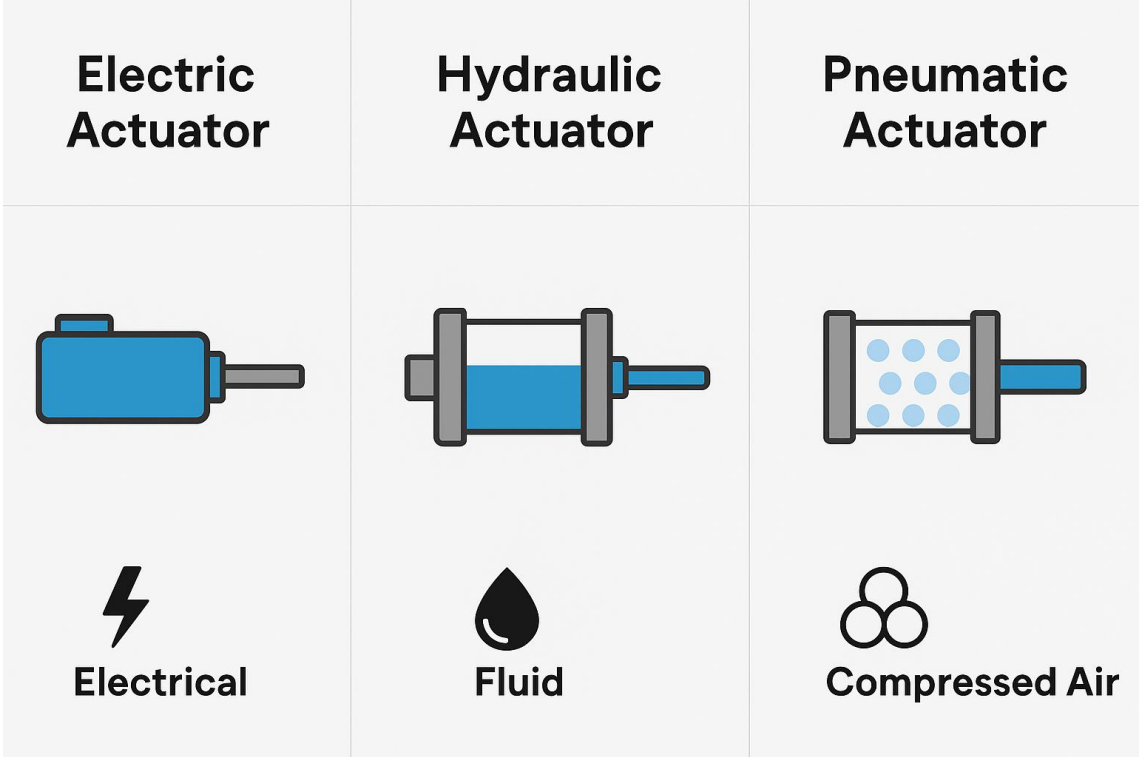
https://www.youtube.com/watch?v=FJBPNJR2QJU



https://youtu.be/XwiX2vv14Qs

Actuators

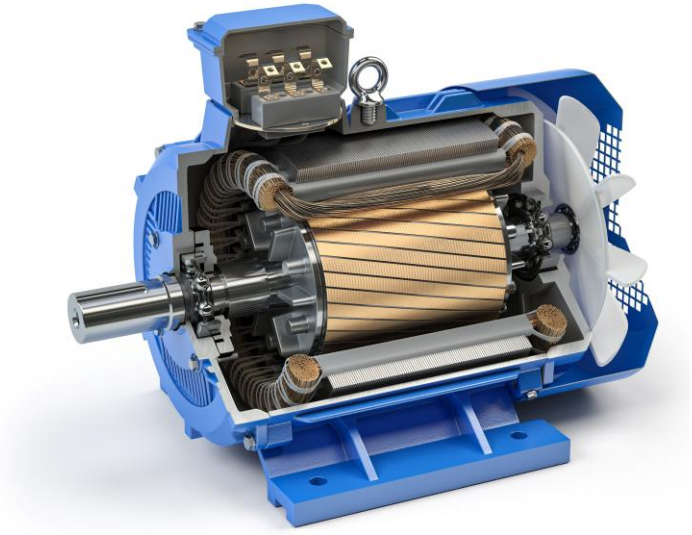
- An **actuator** converts **energy** (usually electrical, hydraulic, or pneumatic) into **mechanical motion** (force or torque)



Hydraulic and Pneumatic Actuators: Electricity → Mechanical rotation (motor) → Fluid/Air pressure (pump) → Motion (actuator)

Actuators

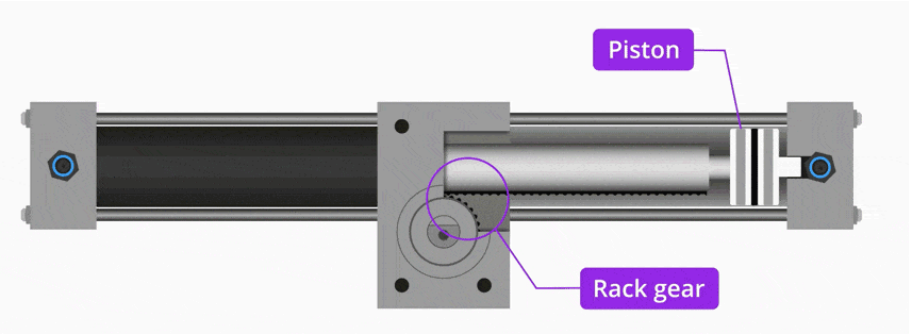
- Electric actuators → Best for precision and control (industrial & collaborative robots).
- Hydraulic actuators → Best for power and stiffness (heavy-duty or dynamic humanoids).
- Pneumatic actuators → Best for speed, simplicity, and soft interaction (grippers, lightweight robots)



Electric actuator



Hydraulic actuator

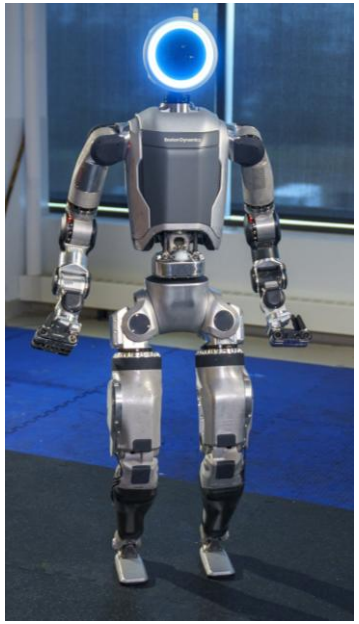


Pneumatic actuator

Boston Dynamics Atlas



Hydraulic Version

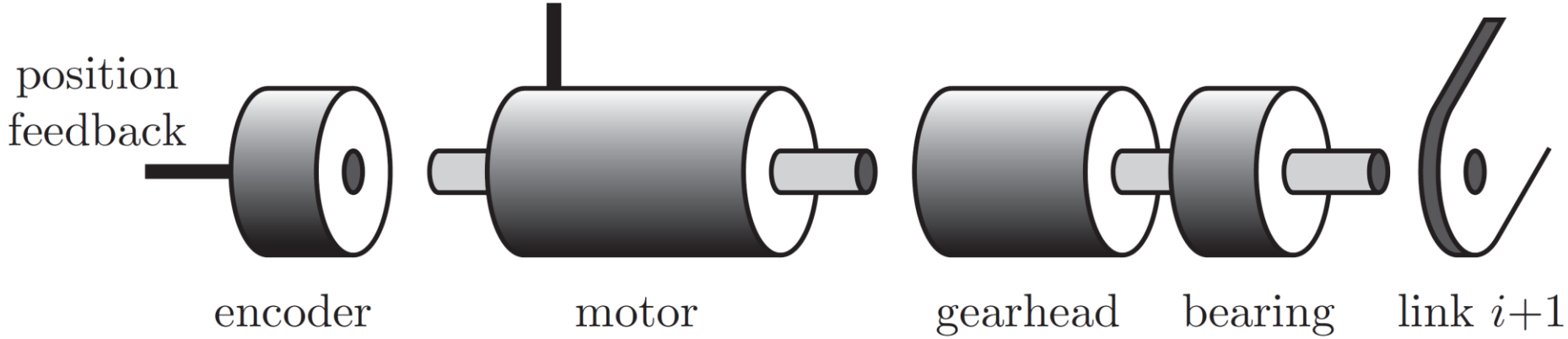


Electric Version

Version	Height	Weight
Atlas (2013, hydraulic)	188 cm (6′2″)	~150 kg (330 lb)
Atlas HD (2016, hydraulic, "next-gen"/battery)	150 cm (4′11″)	80 kg (176 lb)

Actuation with DC Electric Motors

a torque proportional to the current through the motor, this torque command is equivalent to a current command



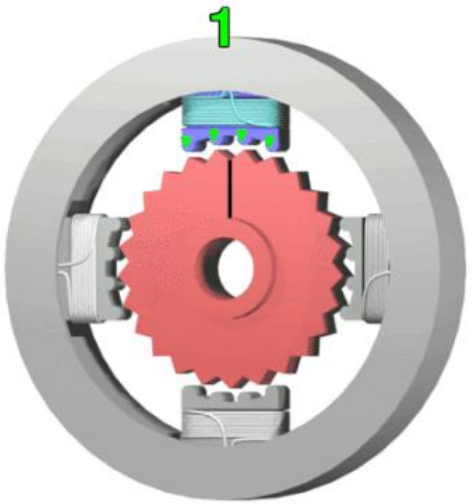
measures the position of the joint

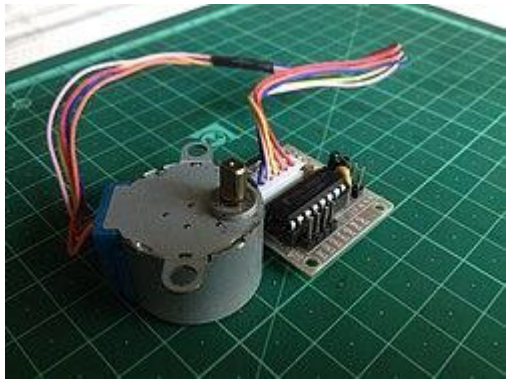
Increase torque
Reduce speed

isolating the gearhead (and motor) from wrench components due to link i + 1

Stepper Motors

- The velocity of a joint is determined directly by the frequency of the pulse train sent to the stepper motor https://en.wikipedia.org/wiki/Stepper_motor





- A **stepper motor** moves in **discrete angular steps** (e.g., 1.8° per step)
- Send **step pulses** at a certain rate (say, 1000 pulses per second)
- Angular velocity

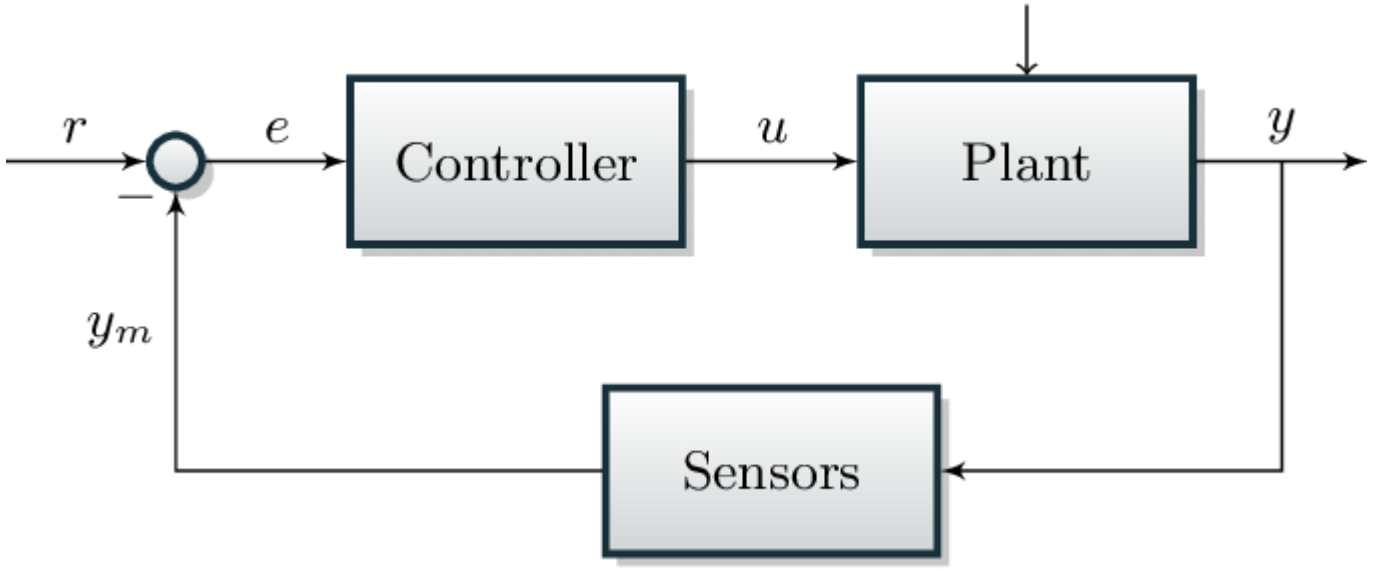
$$\omega = (\text{step angle}) \times (\text{pulse rate})$$

$$\omega = 1.8^\circ \times 1000 = 1800^\circ/s = 5 \text{ revolutions per second}$$

Stepper motors are best for low-speed, precise motion, but not ideal for high-speed or high-torque robotic applications

Using Feedback in Control

- Feedback control
  - Use sensors for position, velocity and force
  - Compare with the desired behavior to compute the control signals



Controlled Dynamics of a Single Joint

- Desired joint position $\theta_d(t)$
- The current joint position $\theta(t)$
- Joint error $\theta_e(t) = \theta_d(t) - \theta(t)$

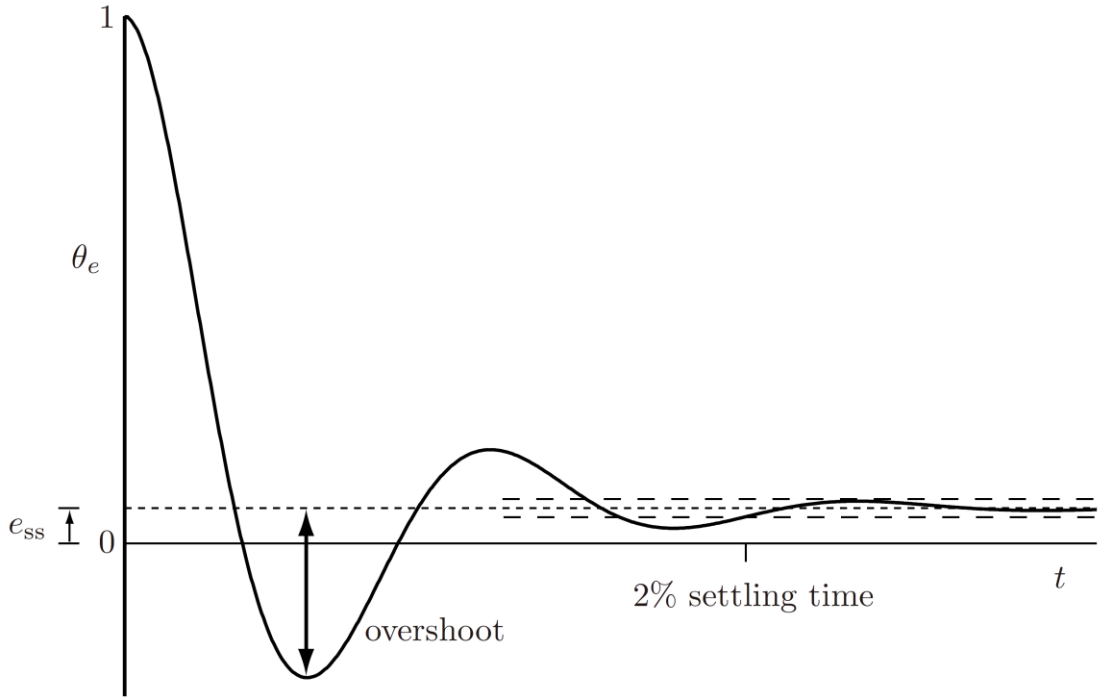
- Error dynamics: the differential equation governing the evolution of the joint error

- Feedback controller: create an error dynamics to make $\theta_e(t)$ become zero or a small value when t increases

Error Response

- How well a controller works?
  - Specify a nonzero initial error $\theta_e(0)$ and see how the controller reduces the error

- Error response $\theta_e(t), t > 0$
  - Initial conditions $\theta_e(0) = 1$
    $\dot{\theta}_e(0) = \ddot{\theta}_e(0) = \cdots = 0$
  - Steady-state error $\theta_e(t)$ as $t \to \infty$

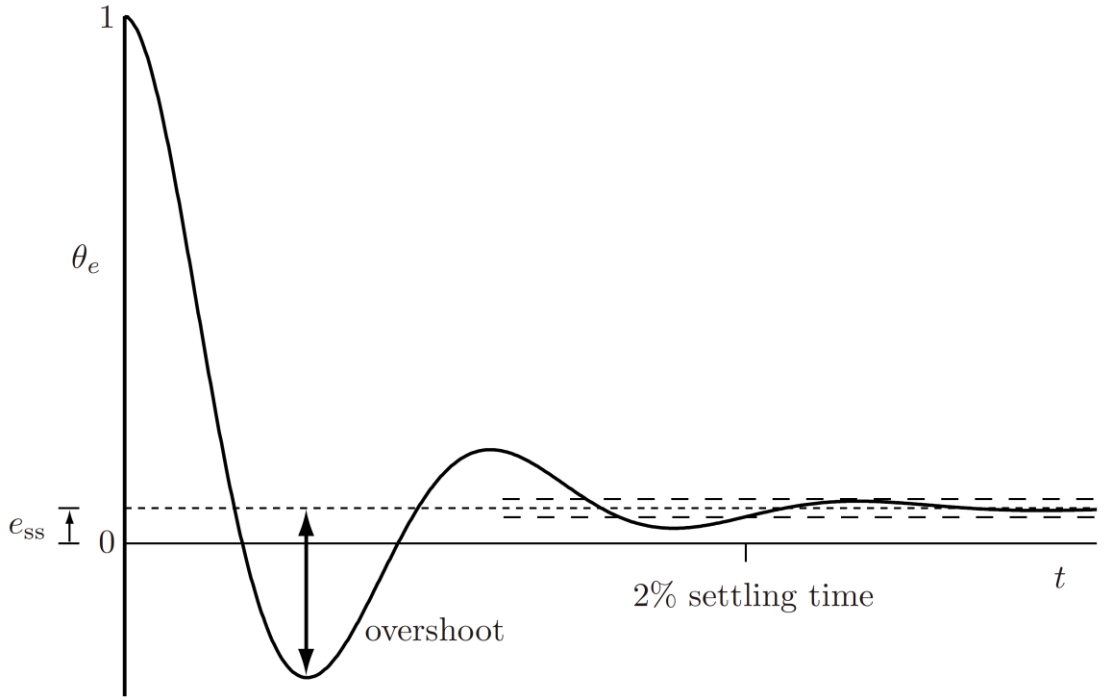


Error Response

- (2%) Settling time: first time T such that $|\theta_e(t) - e_{\text{ss}}| \leq 0.02(\theta_e(0) - e_{\text{ss}})$ for all $t \geq T$

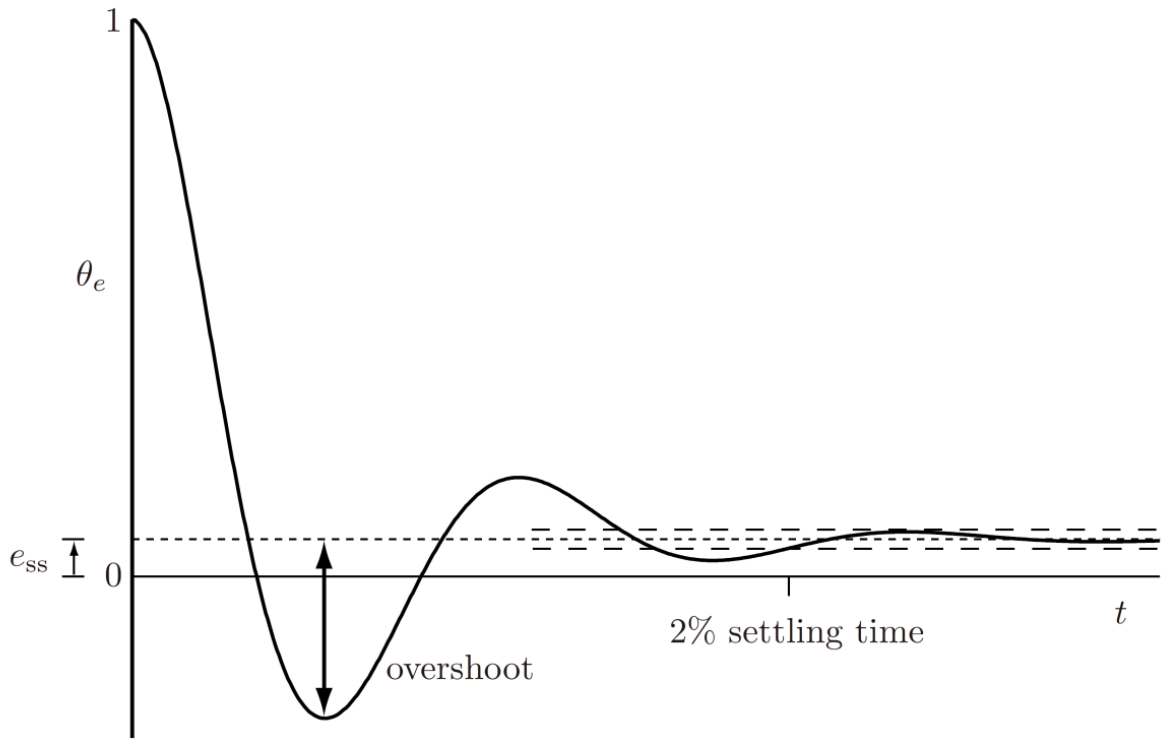
- Overshoot

$$\text{overshoot} = \left| \frac{\theta_{e,\min} - e_{\text{ss}}}{\theta_e(0) - e_{\text{ss}}} \right| \times 100\%$$



Error Response

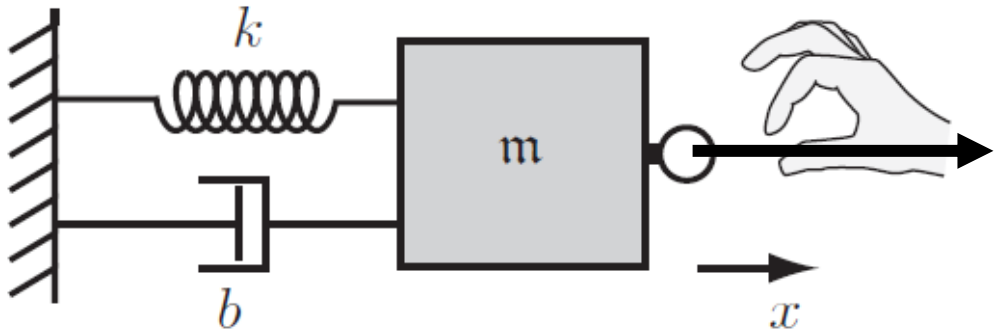
- A good error response
  - steady-state error Little or no
  - overshoot Little or no
  - 2% settling time A short



Mass-Spring-Damper Dynamics

Spring force $-kx$

k= **spring constant**, or **stiffness**



External force f

Damping force $-b\dot{x}$

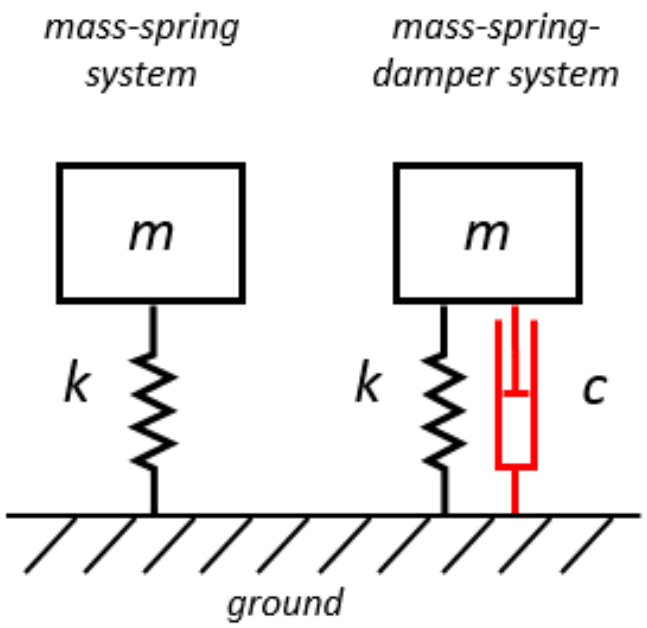
Newton's second law

$$f - kx - b\dot{x} = m\ddot{x}$$

$$m\ddot{x} + b\dot{x} + kx = f$$

Mass-Spring-Damper Dynamics

- Damper: **resists motion** and **dissipates energy** — like friction or air resistance (slowing it down over time)

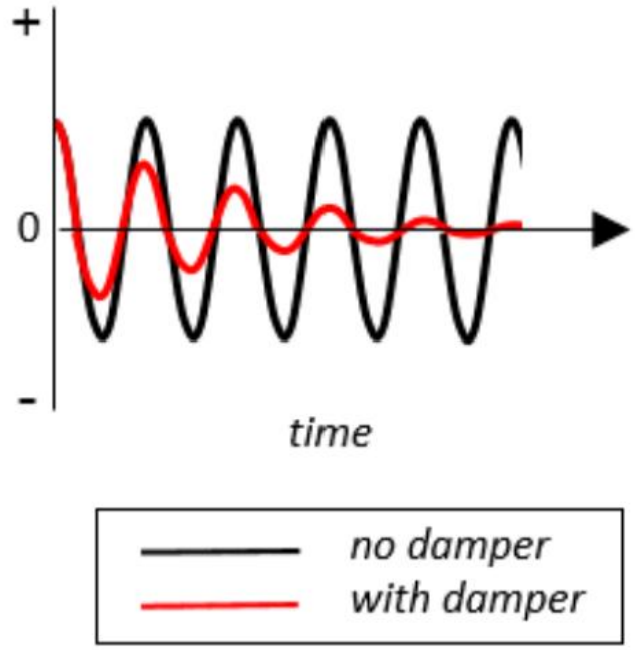


Mass-Spring-Damper Dynamics

$$m\ddot{x} + b\dot{x} + kx = f$$

- Frequency = how many complete cycles (back-and-forth motions) happen per second
  - Unit: Hertz (Hz), cycles per second
- Natural frequency: frequency the system would oscillate at if there were no damping

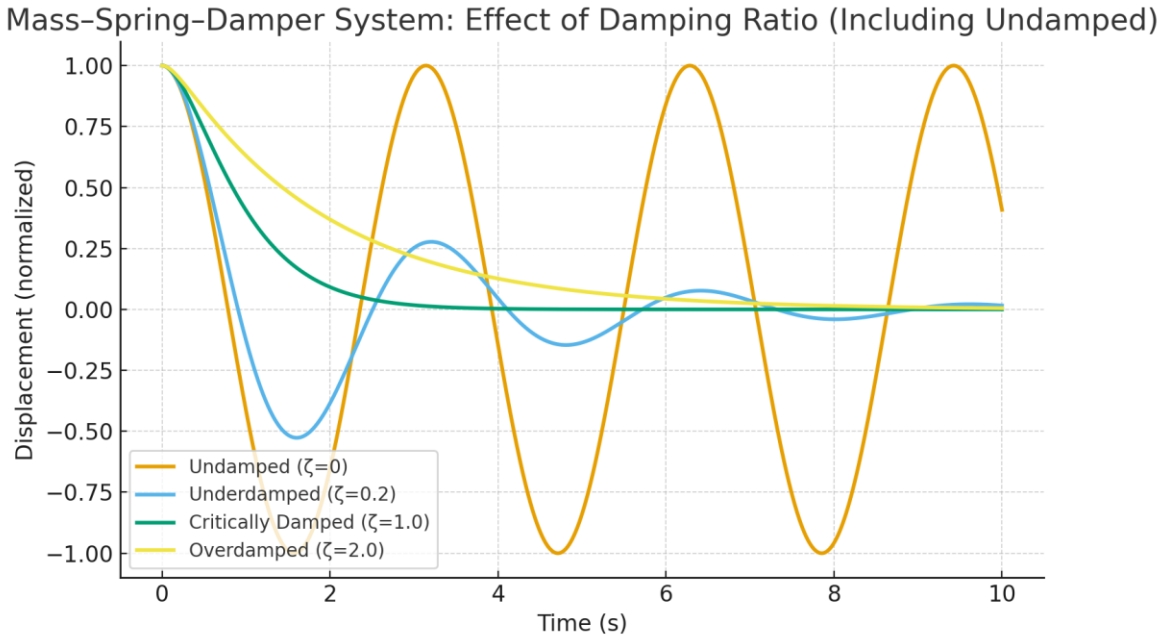
radians per second $\omega_n = \sqrt{\frac{k}{m}}$ Hertz $f_n = \frac{\omega_n}{2\pi}$



- Damped Natural Frequency ω_d

$\omega_d = \omega_n\sqrt{1-\zeta^2}$ Damping ratio $\zeta = \frac{b}{2\sqrt{km}}$

Mass-Spring-Damper Dynamics $m\ddot{x} + b\dot{x} + kx = f$



$$\zeta = \frac{b}{2\sqrt{km}}$$

Damping Ratio (ζ)	Behavior	Motion	Damped Frequency
ζ = 0	Undamped	Pure oscillation	ω_n
0 < ζ < 1	Underdamped	Oscillates and decays	$\omega_n\sqrt{(1-\zeta^2)}$
ζ = 1	Critically damped	No oscillation, fastest return	0
ζ > 1	Overdamped	No oscillation, slow return	*Imaginary* (no real ω_n)

Summary

- Robot control
- Actuators
- Error dynamics
- Mass-Spring-Damper Dynamics

Further Reading

- Chapter 11 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017.